Assignment II

SDG - Based Question - CO_2

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Course code :- ITA0612

Course Name :- Machine Learning.

1) Design a single -layer Perceptron to classify water samples as safe or unsafe.

A single - layer Perceptron classifies water as safe or unsafe using PH, turbidity and dissolved oxygen.

Inputs:-

→ x_1 : PH level
→ x_2 : Turbidity
→ x_3 : Dissolved oxygen.

Net input:-

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Activation function:-

$$y = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

Here, 1 represents safe water and 0 represents unsafe water.

Weight update rule:-

$$w_i(new) = w_i(old) + \eta(t - y)x_i$$

The Perceptron updates weights its Prediction is in-correct. It helps monitor water quality and supports SDG 6.

2) Explain the Perceptron learning rule for classifying soil as fertile or degraded.

3)

A Perceptron classifies soil using nitrogen, Phosphorus and Potassium content.

Net input:-

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Output is 1 for fertile soil and 0 for degraded soil.

learning rule:-

$$e = t - y$$

$$w_i(new) = w_i(old) + \eta e x_i$$

$$b(new) = b(old) + \eta e$$

Convergence conditions:-

1) Training data must be linearly separable.
2) The learning rate must be Positive.
3) Training samples are repeatedly presented.
4) For linearly separable data, the algorithm converges after a finite number of mistakes.

This helps identify soil fertility and supports SDG 2, zero Hunger.

3) Why can't a single Perceptron classify energy Poverty with a non-linear boundary?

A single-layer Perceptron can solve only linearly separable Problems.

Limitations:-

1) It creates only a straight-line decision boundary.
2) It cannot learn complex, non-linear relationships.
3) Energy Poverty depends on income, energy costs, and household requirements.
4) It may Produce incorrect classifications when these factors interact.
5) A multilayer Perceptron can solve such Problems using hidden layers and non-linear activation functions.

Example:- A household with moderate income but very high electricity expenses may still experience energy Poverty.

Thus, multilayer networks can improve energy-Poverty detection and support - SDG 7 Affordable and clean Energy.

4) Design a Perceptron to Predict malnutrition risk. Show two iterations of weight updates.

A Perceptron Predicts malnutrition risk using BMI-related risk and dietary deficiency,

let x_1 and x_2 be the inputs.

$$z = w_1x_1 + w_2x_2 + b$$

Assume

$$w_1 = 0.2,\ w_2 = 0.2,\ b = -0.1$$

learning rate $\eta = 0.1$ and target $t = 1$

Iteration 1:- $x_1 = 1,\ x_2 = 1$

$$z = 0.2 + 0.2 - 0.1 = 0.3$$

Prediction $y = 1$ so error $e = t - y = 0$

No weight update is required.

Iteration 2:- $x_1 = 0;\ x_2 = 1$

$$z = 0 + 0.2 - 0.1 = 0.1$$

Prediction $y = 1$ So error $e = 0$ weights remain unchanged.

The model Predicts whether a Person is at risk of malnutrition and supports SDG 2. Zero Hunger.

7) Explain the limitations of Perceptrons in air-quality classification.

A single Perceptron classifiers air quality as safe or hazardous using Pollutant levels.

Limitations:-

1. It can learn only linear decision boundaries.
2. Pollutants such as PM2.5, nitrogen dioxide and ozone may interact non linearly.
3. It cannot represent complex relationships between Pollutants.
4. It may misclassify air quality when several Pollutants interact.
5. A multilayer Perceptron can learn Complex Patterns using hidden layers and non-linear activation functions.

Example:- A combination of high PM2.5 and nitrogen dioxide levels may create hazardous air quality.

Thus, neural networks can improve air-quality classification and support SDG 11, sustainable cities and communities.